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SUPERSEDING
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MILITARY SPECIFICATION

SENSOR GROUP, GRID WIRE, DT-545/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the grid wire sensor group (hereinafter referred to as "GWS", and the grid wire installation tool (ITL). The GWS is a break wire sensing device used to detect forced entry through a wall, floor, ceiling or door into an area which has been secured (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Reasarch and Development Command, ATTN: DRDME-DS Fort Belvoir, VA 22060, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E3400	- Sensor Group, Grid Wire, DT-545/FSS-9(V).
TA13220E3401	- Tool, Grid Wire Installation.
TA13220E3402	- Sensor Assembly, Grid Wire.
C13220E3406	- Wire, Grid.
B13220E3407	- Staple.
TA13220E3436	- Sensor, Grid Wire.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The GWS (TA13220E3400) shall be as specified herein, and in accordance with applicable drawings specified below. When specified (see 6.2), the GWS, ITL, and major components listed below shall be furnished separately.

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3.1.1 Major components. The major components of the grid wire sensor (TA1322E3436) shall be:

(a) Junction box assembly - TA13220E3402 (see 3.6.1, 3.6.2, 3.6.3).
(b) Wire, Grid - C13220E3406 see 3.6.5.

3.1.2 ITL major components. The major components of the grid wire installation tool shall be:

(a) Tool, grid wire installation - TA13220E3401 (see 3.6.4).
(b) Staples - B13220E3407.

3.1.3 Function. The GWS detects penetration through wall, floors, ceilings, doors and other barriers by a "break wire" method. Any penetration through the barrier in excess of a 4-inch square area results in a break in one or more of the grid wires. The sensor does not produce an intrusion alarm by any means except a physical break in the electrical continuity of the circuit. A break in the electrical continuity of the circuit will result in an alarm being transmitted to, and recognized by, a control unit.

3.1.4 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the GWS or major components (when furnished separately). Any data (e.g. shop drawings, layouts, flow sheets processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the preproduction item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more GWS's or major components (when furnished separately) as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce GWS's or major components (when furnished separately) that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and unless

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otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more GWS's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.5 Environmental conditions. The grid wire sensor assembly shall withstand individual or probable combinations of the environmental conditions specified herein without mechanical or electrical damage or performance degradation, when tested in accordance with 4.6.2.

3.5.1 Operational.

(a) Temperature range of -20° F to +150° F.
(b) Humidity up to 95 percent relative.

3.5.2 Nonoperational and storage.

(a) Temperature range of -30° F to +165° F.
(b) Humidity up to 95 percent relative.
(c) Shock: 20 times the acceleration due to gravity (g's), 11 milliseconds (ms) duration and bench handling.
(d) Vibration: transportation conditions.

3.6 Detailed technical requirements.

3.6.1 Grid wire sensor assembly. An internal tamper switch shall be provided which is activated when the cover is in place and shall open when the cover is moved not less than 0.125 inch or more than 0.250 inch from the normally closed position and before the cover flanges disengage from the housing or base plate. A pry-off tamper switch shall be provided on the back of the enclosure to open contact upon removal of the enclosure 0.125 to 0.250 inch from its mounted position.

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3.6.2 Terminal board connnections. The terminal board (TB) shall be marked with identifying terminal numbers as follows:

TB-	Function
1	Horizontal grid
2	Horizontal grid
3	Vertical grid
4	Vertical grid
5	Tamper
6	Tamper Return
7	Enclosure ground

3.6.3 Alarm conditions. The alarm output conditions across the sensor alarm contacts TB-5 and TB-6 shall be as follows:

(a) No alarm: 0 to 100 ohms.
(b) Alarm: more than 100 kilohms (k ohms).

3.6.4 Grid wire installation tool. The ITL shall automatically dispense the grid wire and be manually operated to fasten the wire to the foundation board with staple-type fasteners. The fasteners shall be such that the mounting surface shall be penetrated to a depth of 0.250 $\pm$ 0.062 inch. The installation tool shall have a magazine capable of dispensing a minimum of 125 fasteners before requiring a refill. The ITL shall be manually refilled by inserting the clips of fasteners into the magazine. The ITL shall develop a tension of not less than 1 pound and not greater than 4 pounds on the wire during application. The wire shall be dispensed through a positive guide system such that no alinement is necessary to assure that the staple will capitivate the wire. On soft material walls, the staple shall not cut into the wire. The ITL shall be capable of dispensing a minimum of 600 feet of grid wire. The weight of the installation tool, including wire and fasteners, shall be not greater than 4 pounds.

3.6.5 Grid wire. The grid wire shall be No. 26 AWG enamel-coated, solid copper wire. The tensile strength of the wire shall be 35,000 to 45,000 pounds per square inch. The grid wire shall be wound on a spool containing at least 600 feet of grid wire.

3.6.6 Staples. The staples shall be assembled into frangible clips of approximately 125 staples suitable for loading into the ITL.

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3.7 Identification marking. The GWS and major components (when furnished separately), shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Workmanship. Workmanship shall be in accordance with the requirements of MIL-STD-454, Requirement 9 except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.9.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and, standards, and drawings as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part

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shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.4 and 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction GWS's and major components (when furnished separately) shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The GWS shall be subjected to the tests marked "X" in Column 1 of Table I (see 4.6.2). Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.1 Major components furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in Column 1 of Table II or Table III as applicable. Failure of any tests shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production GWS's will be selected at random by the Government from the GWS's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification.

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The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production GWS shall not exclude the remaining GWS's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production GWS to meet any requirement specified herein during and as a result of the examinations and tests specified in 4.4 shall be cause for rejection of the initial production GWS's and shall be cause for refusal by the Government to continue acceptance of production GWS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on GWS's previously accepted and produced under the contractor. Any deficiencies found as a result of the inital production inspection will be considered prima facie evidence that all GWS's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II. Sampling for test shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the major defects specified in 4.6.1. AQL shall be 2.5 percent defective.

4.5.3 Tests.

4.5.3.1 Samples. GWS's selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 2 of Table II. AQL shall be 2.5 percent defective.

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4.5.2.1.2 Major components. Major components (when furnished separately) selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 2 of Table II or Table III as applicable. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The GWS's and any major component (when furnished separately), shall be examined as specified herein for the following major defects.

101. Any part (or component) not in accordance with the SQAP requirements shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.
104. Interchangeability not as specified.
105. Identification or special marking missing or illegible.
106. Workmanship including wiring and soldering not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Test equipment shall be calibrated in accordance with MIL-C-45662.

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4.6.2.2 Test schedule.

TABLE I. Test schedule for Sensor group, grid wire.

Preproduction	Initial production	Quality Conformance		Test	Requirement paragraph	Test paragraph
		Individual	Sample			
1	2	3	4	5	6	7
				Functional Test:		
X	X	X	X	Sensor assembly, operation.	3.6.1, 3.6.2, 3.6.3	4.6.2.3.1
X	X	-	X	Installation tool.	3.6.4	4.6.2.3.2
X	X	-	X	Wire strength, break test.	3.6.5	4.6.2.3.3
X 1/	X 1/	-	-	Shock test.	3.5.2	4.6.2.4.2
X 1/	X 1/	-	X 1/	Humidity.	3.5.1, 3.5.2	4.6.2.4.3
X 1/	X 1/	-	X 1/	Low temperature.	3.5.1, 3.5.2	4.6.2.4.4
X 1/	X 1/	-	X 1/	High temperature.	3.5.1, 3.5.2	4.6.2.4.5
X 1/	X 1/	-	-	Vibration.	3.5.2	4.6.2.4.6

1/ Test only the grid wire sensor assembly, see 3.1.1(a)

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TABLE II. Test schedule for grid wire sensor furnished separately.

Preproduction	Quality Conformance Sample	Test	Requirement paragraph	Test paragraph
1	2	3	4	5
		Functional test:		
X	X	Operation.	3.6.1, 3.6.2	4.6.2.4.1
X	-	Shock test.	3.5.2	4.6.2.5.2
X	X	Humidity.	3.5.1, 3.5.2	4.6.2.5.3
X	X	Low temperature	3.5.1, 3.5.2	4.6.2.5.4
X	X	High temperature	3.5.1, 3.5.2	4.6.2.5.4
X	-	Vibration.	3.5.2	4.6.2.5.6

TABLE III. Test schedule for grid wire installation

Preproduction	Quality Conformance Sample	Test	Requirement paragraph	Test paragraph
1	2	3	4	5
X	X	Installation tool.	3.6.4	4.6.2.4.2
X	X	wire strength.	3.6.5	4.6.2.4.3

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4.6.2.3 Test methods. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with test methods in the Appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.2.4 Functional tests.

4.6.2.4.1 Sensor assembly operation. The grid wire sensor assembly, shall be tested in accordance with Appendix Test Method 1, Nonconformance to 3.6.1, 3.6.2, and 3.6.3 shall constitute failure of this test.

4.6.2.4.2 Installation tool. The ITL shall be tested in accordance with Appendix Test Method 2. Nonconformance to 3.6.4 shall constitute failure of this test.

4.6.2.4.3 Wire strength - break test. The wire shall be tested in accordance with Appendix Test Method 3. Nonconformance to 3.6.5 shall constitute failure of the test.

4.6.2.5 Environmental test. The grid wire sensor assembly shall meet all specified requirements when subjected to the following sequence of environmental tests of MIL-STD-810 in accordance with the following conditions.

4.6.2.5.1 Functional test. Test in accordance with test specified in 4.6.3.1.1.

4.6.2.5.2 Shock test. Conduct the following tests:

(a) Basic design: Test Method 516.2, Procedure I. Equipment not operating, sawtooth shock pulse, peak value 20 g's, duration 11 ms.
(b) Bench handling: Test Method 516.2, Procedure V.

4.6.2.5.3 Humidity test. Conduct Test Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Perform measurements at the completion of each cycle.
(b) During Steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the realtive humidity shall be maintained at 92 percent or greater.

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(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. The GWS or major component (when furnished separately) may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the GWS or major component (when furnished separately) to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test. The cover of the sensor assembly may be removed for the last functional test. Wipe excess moisture from the cover prior to removing cover.

4.6.2.5.4 Low temperature test. Conduct low temperature Test Method 502.1, Procedure I, storage temperature -30° F for not less than 48 hours, equipment operating temperature -20° F. Perform functional test at operating temperature.

4.6.2.5.5 High temperature test. Conduct high temperature Test Method 501.1, Procedure II, except that the nonoperating temperature shall be +165° F; the operating temperature shall be +150° F. Perform functional test at the operating temperature.

4.6.2.5.6 Vibration test. Conduct Test Method 514.2, Procedure X, Curve AX, equipmennt nonoperating.

4.6.2.5.7 Failure criteria. Nonconformance to 3.5 shall constitute failure of the test.

4.7 Inspection comparison. The Government may select GWS's at any time during the contract production period and subject these GWS's to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on GWS's selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production GWS's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison GWS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the inspection comparison GWS and shall be cause for refusal by the Government to continue acceptance of production GWS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on GWS's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all GWS's accepted prior to the completion of the inspection comprison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for Level A, or Level B. Each incorrect material, method, or container shall be considered one defect.
108.	Components not preserved by the applicable procedure for Level A, or Level B.
109.	Boxed contents not cushioned to prevent movement or damage as specified for Level A, or Level B.
110.	Box not closed and sealed by method specified for Level A, or Level B.
111.	Components not packed by the applicable procedure for Level A, or Level B.
112.	Strapping not as specified for Level A.
113.	Quantities packed together exceed the size or weight limitations of the box for Level A, or Level B.
114.	Marking missing, illegible, incorrect, or incomplete for Level A, or Level B.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, or Level B, or Commercial as specified.

5.1.1 Level A. The GWS components shall be preserved by one of the following procedure, depending upon procurement of complete GWS's or procurement of components furnished separately.

5.1.1.1 Preservation of complete GWS's. The components comprising a complete GWS shall be preserved in accordance with MIL-P-116 as follows: The grid wire sensor assembly shall be preserved Method II c; the grid wire installation tool and box of staples shall be preserved Method IC-2, using the contractor's carton or a box conforming to PPP-B-566; Variety 1, style, type, and class optional, or PPP-B-676, Variety 1, style, type, and class optional; all other components shall be individually preserved Method IC-1 or IC-3. The grid wire sensor assembly shall then be individually placed in a close-fitting box conforming to PPP-B-636, W5c, or W6c, style optional and cushioned within the box to prevent movement or damage. All other components required for one complete GWS shall be placed together in a box and cushioned as specified herein for the grid wire sensor assembly.

5.1.1.2 Major components furnished separately.

5.1.1.2.1 Grid wire sensor assemblies. When only grid wire sensor assemblies are procured, they shall be individually preserved and placed in a box as specified in 5.1.1.1.

5.1.1.2.2 Installation tool, grid wire, staples, sandpaper, crimping tool, and terminal lugs. When these components are furnished separately, they shall be individually preserved as specified in 5.1.1.1 and then components of like description placed together in a close-fitting box and cushioned as specified in 5.1.1.1 in quantities not to exceed the weight or size limitations of the box.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A, except Method IA-8 shall be used in place of Method II c.

5.1.3 Commercial. Each complete GWS or separately procured GWS component shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, or Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. The GWS components, preserved as specified in 5.1, shall be packed by one of the following procedures depending upon procurement of complete GWS's or procurement of components separately.

5.2.1.1 Packing of complete GWS. The components comprising a complete GWS shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.1.2 Packing of major components furnished separately. GWS components of like description shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Complete GWS's or individually furnished components shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

6.1 Intended use. The GWS covered by this specification is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of GWS's, ITL's, and major components to be furnished separately (see 3.1).
(c) Time frame required for submission of the preproduction model and number of grid wire sensor assemblies, or major components (when furnished separately) required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examinations and tests. When the Government will conduct some but not all of the preproduction examinations and tests, the contracting officer should specify which examinations and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of GWS's to be furnished (see 3.3).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(g) When other than Finish B strapping is required (see 5.2.1.1).

6.3 Preproduction model. Any changes or deviations of production GWS's or major components furnished separately, from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish GWS's or major components (furnished separately) conforming to this specification (see 3.2).

6.4 Definitions. The following definitions shall apply throughout this specification:

6.4.1 False alarm. An alarm generated by stimuli other than those for which the GWS is designed to alarm.

6.4.2 Alarm signal.

6.4.2.1 Tamper alarm. Alarm resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component, or physical change which impairs the mechanical integrity of the GWS or major component (furnished separately). Evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components or

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physical condition resulting in noncompliance of the sensor assembly with the provisions of this specification shall be cause for rejection.

6.4.4 Supplementary quality assurance provisions. A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Supplementary quality assurance provision (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each GWS or major component.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the GWS.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME
Air Force - 99

Preparing activity:
Army - ME

Review activity:
DLA - GS

Project 6350-0062

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APPENDIX

TEST METHODS

FOR

SENSOR GROUP, GRID WIRE

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether sensors procured under this specification conform to the requirements specified.

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TEST METHOD 1

FUNCTIONAL TEST

1. Test apparatus:

(a) Ohmmeter - Darcy Model 330, or equivalent.
(b) Ruler.

2. Test procedure:

(a) Grid wire sensor assembly.

(1) Remove sensor assembly cover and connect ohmmeter between TB-5 and TB-6.

Note: Insert ohmmeter leads through conduit entry port to facilitate installation of sensor assembly cover.

(2) Place sensor assembly on a work bench with its bottom or mounting surface down and securely place cover on sensor assembly. Ohmmeter indication should not exceed 100 ohms.
(3) Lift sensor assembly from work bench (lift from one corner) until ohmmeter indicates more than 100k ohms. Distance of travel should between 0.125 and 0.250 inch.
(4) Lower sensor assembly to work bench surface. Ohmmeter indicator should not exceed 100 ohms.
(5) Lift sensor assembly cover (lift from edge with notch for conduit) while holding the sensor assembly securely on work bench until ohmmeter indicates more than 100k ohms. Distance of travel should between 0.125 and 0.250 inch.
(6) Disconnect ohmmeter test leads and install cover and cover screws.

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TEST METHOD 2

GRID WIRE INSTALLATION TOOL

1. Test apparatus:

(a) Wire - Belden 8053, or equivalent.
(b) Sample construction lumber and sheetrock.
(c) Spring scale.

2. Test procedure:

(a) Assemble a spool of Belden 8053 wire to the installation tool.
(b) Install one length of staples into the installation tool.
(c) Adjust tension on installation tool for proper penetration of 0.250 ± 0.0625 inch. Drive five staples into lumber sample and five staples into a sheet of sheetrock. The thickness of the wood and the sheetrock shall be not less than 1/2 inch.
(d) Attach spring scale to the free end of the wire after it is threaded through the wire guides on the installation tool.
(e) Hold the installation tool still and pull on the spring scale until the reel turns. Apply pull parallel to the base of the installation tool. Pull required to turn the reel shall be not less than 1 pound nor greater than 4 pounds.

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TEST METHOD 3

WIRE STRENGTH - BREAK TEST

1. Test apparatus:

   (a) Spring scale.
   (b) Stationary anchor.

   Note: Commercial wire test assembly may be used.

2. Test Procedure:

   (a) Cut a piece of 26-gage grid wire 56 inches long (minimum) for use as a test specimen.
   (b) Connect test wire between anchor and spring scale.

   Note: Ensure that there are no kinks in the wire before subjecting the wire to break strength testing.

   (c) While observing the scale, apply a constant, steadily increasing load parallel to the length of the wire until the test wire breaks. The test wire shall break at 7.0 to 9.0 pounds tension.

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-52861B Sensor Group, Grid Wire, DT-545/FSS-9(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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